

Documentation for ETSI GS QKD 004 push mode specification, QKD server

Documentation for API Endpoints

All URIs are relative to `https://qkd_server/api/v1/kms/etsi004`

Class	Method	HTTP request	Description
<i>Etsi004Api</i>	getKey	GET /get-key	start pushed key retrieval process
<i>Etsi004Api</i>	postOpenConnect	POST /open-connect	open key stream
<i>Etsi004Api</i>	putClose	PUT /close	close the key stream

Documentation for Models

- `get_key_205_response`
- `get_key_request`
- `metadata`
- `post_open_connect_201_response`
- `post_open_connect_request`
- `put_close_204_response`
- `put_close_request`
- `qos`

Documentation for Authorization

All endpoints do not require authorization. # Etsi004Api

All URIs are relative to `https://qkd_server/api/v1/kms/etsi004`

Method	HTTP request	Description
getKey	GET /get-key	start pushed key retrieval process
postOpenConnect	POST /open-connect	open key stream
putClose	PUT /close	close the key stream

getKey > `get_key_205_response` `getKey(get_key_request)`

start pushed key retrieval process

The ``get-key`` message is the starting point after which the QKD device can start pushing keys

Parameters

Name	Type	Description	Notes
<code>get_key_request</code>	<code>get_key_request</code>		

Return type

`get_key_205_response`

Authorization

No authorization required

HTTP request headers

- **Content-Type:** `application/json`
- **Accept:** `application/json`

postOpenConnect > `post_open_connect_201_response` `postOpenConnect(post_open_connect_request)`

open key stream

The ``open-connect`` message connects a KMS to the QKD Device. Conforming to the specification,

Parameters

Name	Type	Description	Notes
<code>post_open_connect_request</code>	<code>post_open_connect_request</code>		

Return type

`post_open_connect_201_response`

Authorization

No authorization required

HTTP request headers

- **Content-Type:** application/json
- **Accept:** application/json

```
# putClose > put_close_204_response putClose(put_close_request)
```

close the key stream

The ``close`` message disconnects the KMS from the QKD device.

Parameters

Name	Type	Description	Notes
<code>put_close_request</code>	<code>put_close_request</code>		

Return type

`put_close_204_response`

Authorization

No authorization required

HTTP request headers

- **Content-Type:** application/json
- **Accept:** application/json

get_key_205_response

Properties

Name	Type	Description	Notes
<code>status</code>	<code>Integer</code>		[default to null]
<code>metadata</code>	<code>metadata</code>		[optional] [default to null]

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get_key_request

Properties

Name	Type	Description	Notes
key_stream_id	UUID	corresponding to the one given in ‘open_connect’	[default to null]
index	Integer	increasing index of the key chunk in the key stream. Wrapping at UINT_MAX.	[optional] [default to null]
metadata	metadata		[optional] [default to null]

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metadata

Properties

Name	Type	Description	Notes
size	Integer	size of buffer in characters	[optional] [default to null]
buffer	String	metadata in json format.	[optional] [default to null]

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post_open_connect_201_response

Properties

Name	Type	Description	Notes
source	URI	the source URI of the request.	[default to null]
destination	URI	the destination URI of the request.	[default to null]
key_stream_id	UUID	key stream ID. If not specified it is up to the QKD device to define the UUID	[optional] [default to null]
qos	qos	requested QoS	[default to null]
status	Integer		[optional] [default to null]

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post_open_connect_request

Properties

Name	Type	Description	Notes
source	URI	the source URI of the request.	[default to null]
destination	URI	the destination URI of the request.	[default to null]
key_stream_id	UUID	key stream ID. If not specified it is up to the QKD device to define the UUID	[optional] [default to null]
qos	qos	requested QoS	[default to null]

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put_close_204_response

Properties

Name	Type	Description	Notes
status	Integer		[default to null]

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put_close_request

Properties

Name	Type	Description	Notes
key_stream_id	UUID	corresponding to the one given in ‘open_connect‘	[default to null]

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qos

Properties

Name	Type	Description	Notes
key_chunk_size	Integer	length of key chunks in Byte.	[optional] [default to null]
max_bps	Integer	in bit per second	[optional] [default to null]
min_bps	Integer	in bit per second	[optional] [default to null]
jitter	Integer	in bit per second	[optional] [default to null]
priority	Integer	Meaning up to implementation.	[optional] [default to null]
timeout	Integer	Timeout in msec after which the call will be aborted, returning an error.	[optional] [default to null]
ttl	Integer	Time in seconds after which the keys for this key stream ID shall be erased from the kms key store.	[optional] [default to null]
metadata_mimetype	String	MIME type of the metadata of the ‘get‘ request	[optional] [default to null]

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